



Procedure #18

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FEEDWATER SYSTEM STARTUP PROCEDURE

This procedure assists the operator in performing a thorough check of the boiler feedwater system and condensate return system.

Water Softener System Startup Checks

The system consumes about 20 kilograms of salt per month.

The softeners regenerate AUTOMATICALLY when the integrator on the softener controller reaches zero. After regeneration completes, the integrator resets to 48,000 gallons which is approximately once per month.

Step	Operation	Initial when Complete	Date and Time when Complete
1	Verify the brine tank has the proper level of salt and water. The salt bed should be at one-third of the height of the tank.		
2	Open the water softener supply valve (W-4) and all upstream valves leading to the softener on the domestic water line.		
3	Check that softener outlet valve (C-88) is open.		
4	Check that the softener multi-port valve unit has power. This can be seen if the LCD screen is green. Use "The 3200NXT Water Softener Controller Manual" to troubleshoot any problems.		
5	Check the proper operation of level control valve C-69, which adds makeup to the hotwell tank.		
6	Check the level of the Hotwell using the gauge glass. The level should be at least 50 percent of the gauge glass.		

Chemical Checks			
Step	Operation	Initial when Complete	Date and Time when Complete
7	Check that the feedwater chemical tank is filled and that a proper dosage charge has been made, as per the feedwater treatment manual.		
8	Check the feedwater chemical injection pump is on and set correctly as per the feedwater treatment manual or as per previous log book recorded settings.		
Note: During normal operation, the hotwell tank water heater uses live steam fed directly into the tank sparge pipe. Ensure that the steam supply valve is open (S-67). During operation, the Spence PRV and T14 regulator maintain water temperature between 190 and 200°F. The maximum water temperature should not exceed 200°F to prevent boiling off amines.			
Startup Chemical Batching			
Step	Operation	Initial when Complete	Date and Time when Complete
9	If the Hotwell has been drained or is full of fresh water, a start-up charge as specified in the feedwater treatment manual must be added directly to the condensate return tank.		
Note: Feedwater is to be tested each shift to verify residuals are within limits listed in the feedwater treatment manual.			
Condensate Return Tank and Transfer Pumps Checks			
10	Open the condensate return tank inlet valve C-1		
11	Open the condensate transfer pump discharge valves, C-2 and C-3		
12	Verify that power is on to the condensate transfer pumps.		

Blow Down Tank Checks

Step	Operation	Initial when Complete	Date and Time when Complete
13	Check the blow down tank is ready for service. Close the drains if required.		
14	Open cooling water supply valve S-11.		
15	Check operation of the thermostatically controlled valve downstream from valve S-11.		
16	Open valves to allow continuous blow down and intermittent blow down water to get to the blow down tank.		

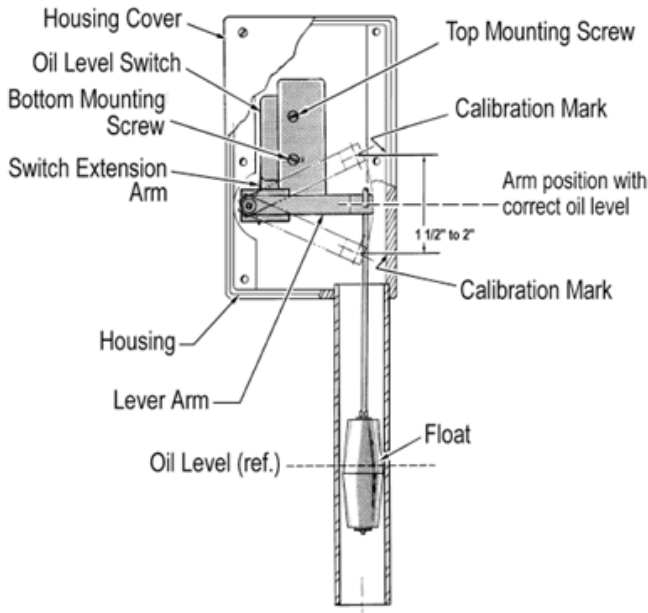
TDS Controller Checks

Step	Operation	Initial when Complete	Date and Time when Complete
17	Check that the TDS controller is online. The LCD screen will indicate the feedwater TDS. The TDS sensor is located in the discharge line of the booster pumps.		

Booster Pumps

Step	Operation	Initial when Complete	Date and Time when Complete
18	Verify that the inlet and discharge valves (S-6, S-7, S-8, and S-9) are open on the booster pumps. <i>(Normal booster pump pressure is around 12 PSig. One booster pump is in service at a time while the other is on standby. Operation is set on the DDC.)</i>		
19	Turn on the power to the booster pumps. The switches are on located on the Hotwell support leg.		

Feedwater Pumps

Step	Operation	Initial when Complete	Date and Time when Complete
20	Open the inlet and discharge valves on the feedwater pump that is in service.		
21	Clayton Boiler Specific - Verify the pump crank case on the Clayton Boiler feedwater pump has the proper oil level. This is done by checking the position of the float operated switch. See diagram below.		
 Figure 4 - Clayton Feedwater Pump Oil Switch			
DANGER: The Clayton Boiler's feedwater pump is directly controlled and powered by the Clayton Boiler's control system. The pump may activate periodically to flow feedwater through the boiler. If the Clayton Boiler is set to "Fill" or "Run", the feedwater pump will automatically start.			

Step	Operation	Initial when Complete	Date and Time when Complete
22	C.B. Boiler Specific - Verify open the valves in the line from the feedwater pump discharge to the boiler. <u>Note: There is an electrically-operated solenoid valve in the line coming from the pump discharge just before the feedwater line ties into the C.B. Boiler.</u>		
23	Make an appropriate log book entry.		
End of Procedure			